ZEISS India Hackathon · Beyond Code 2026

AI/ML for Advanced Frugal Innovation

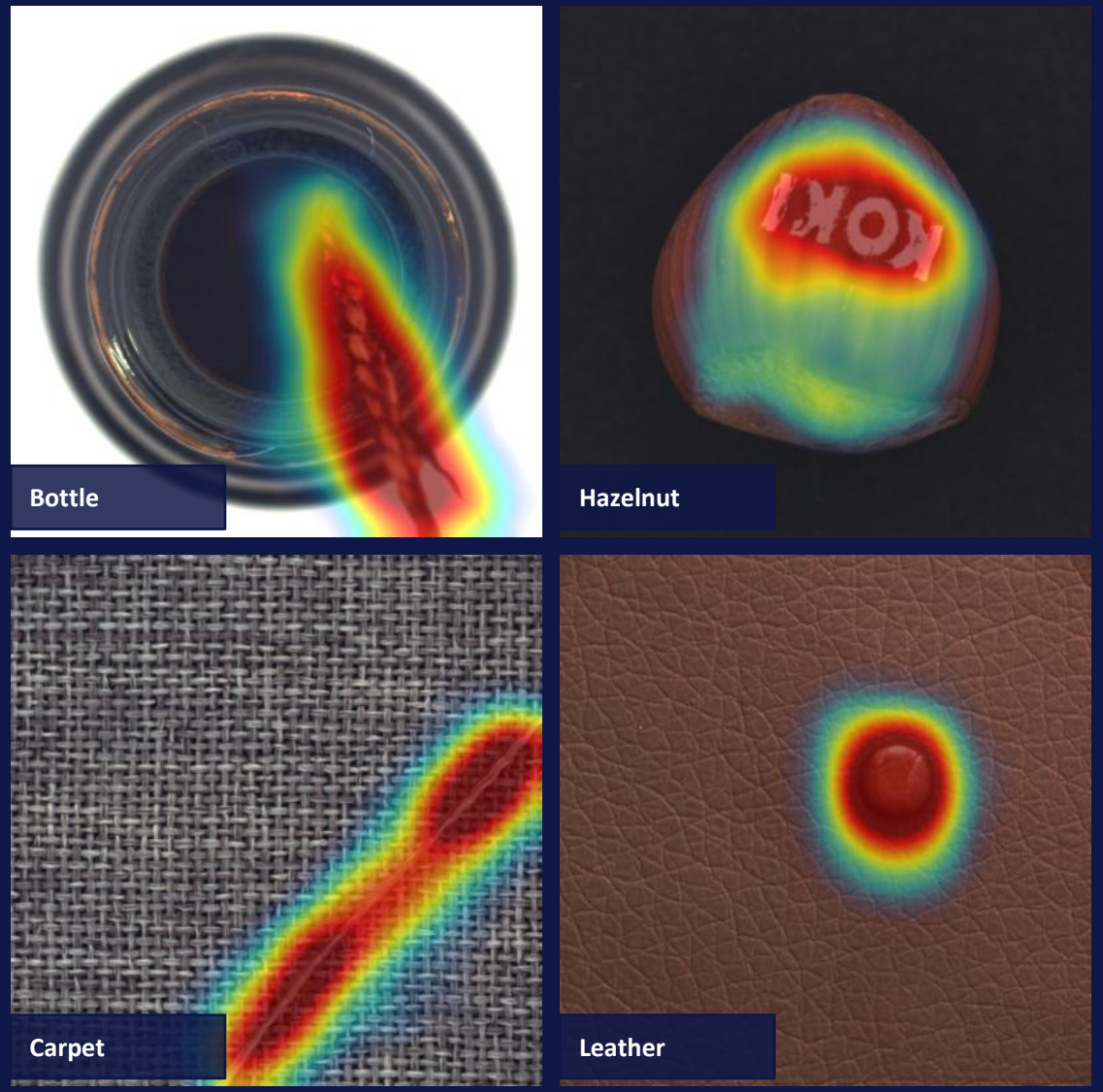
MID-PROGRESS UPDATE

NetraQC

Low-cost surface inspection for small factories, inspired by ZEISS SurfMax

Team Tensor-hive, IIT Kanpur

8 November 2026



MVTec AD test images with NetraQC heatmaps

Our objective, and what we have done so far

Objective An inspection station a small factory can set up from about 20 good parts. It flags surface defects with a heatmap, runs offline on a ₹12k Raspberry Pi kit, and uses ML to choose the cheapest design that still catches defects.

Sprint goal By 15 November: a working offline prototype, tested beyond one product type, with the lighting question measured.

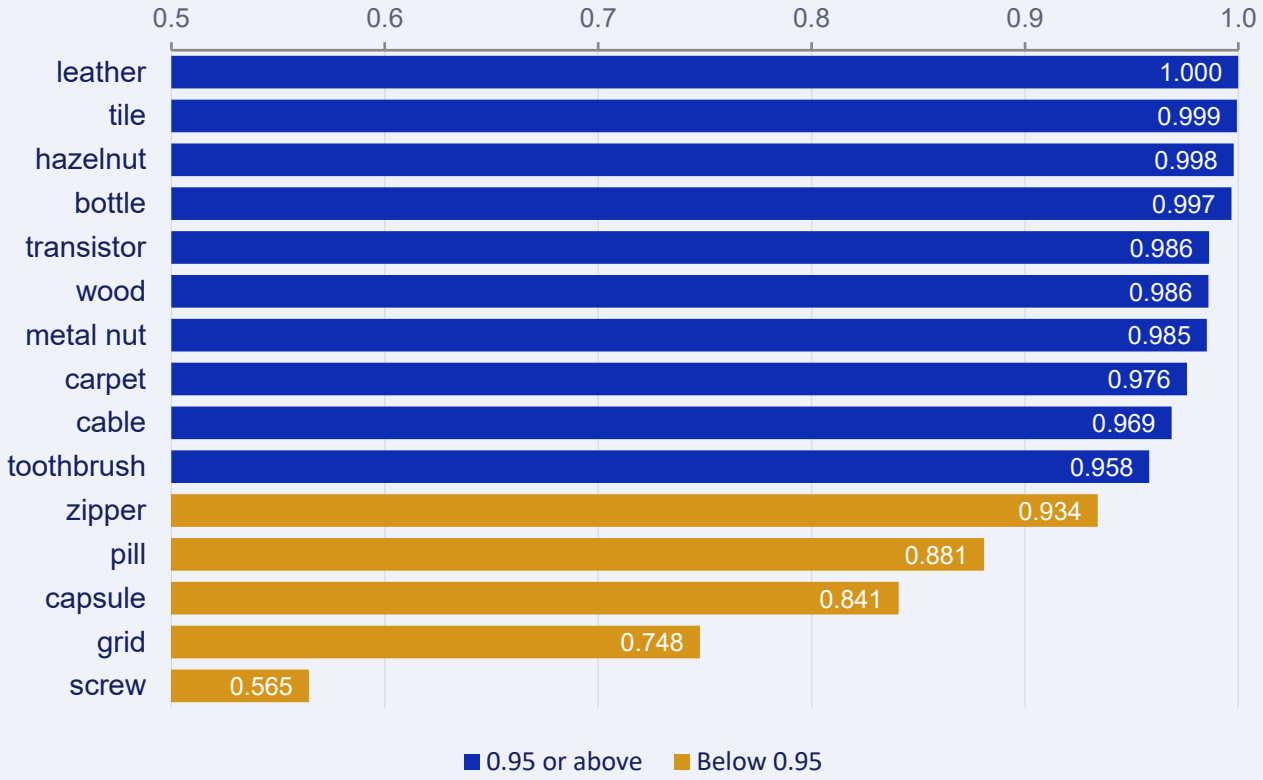
#	DONE IN THE SPRINT	STATUS	WHAT WE FOUND
1	**Compress the detector for edge devices (ONNX, INT8)**	**Done**	11.1 → 2.9 MB and 1.6× faster on 4 CPU threads; AUROC drop at most 0.004 on 4 categories
2	**Check that light adds up, so light combinations can be simulated**	**Done: no**	No: on Eyecandies renders the all-lights shot misses the sum of single lights by 28–57%. Simulation needs linear RAW captures
3	**Compare one shot with several lights**	**Early signal**	Four directional lights: AUROC 0.735 vs. 0.583 for one all-lights shot (cropped, 320 px; 50 samples)
4	**Search for the fewest lights that keep accuracy**	**Early signal**	All 63 combinations scored: 2 directional lights (0.755) beat all 6 together (0.699). Picked on the test samples, so optimistic
5	**Run on all 15 MVTec AD product types, no per-type tuning**	**Done**	Mean image AUROC 0.921; 10 of 15 at 0.95 or above; 923/1258 defects caught

Already built before the sprint: the anomaly detector (metal nut: 0.985 AUROC, 82 of 93 defects caught, 0 of 22 false alarms) and the ML design optimiser (300 designs; a 131 MB model bought no accuracy over our 13.5 MB one). Code: github.com/vatsal3512/zeiss-netraqc

One fixed design on 15 product types: mean AUROC 0.921

Same model, settings and threshold rule for every type; each set up from 20 defect-free images.

Image AUROC by product type



923 / 1258

defective images caught at the threshold

53 / 467

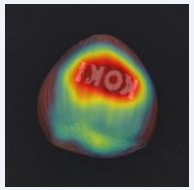
defect-free images wrongly flagged

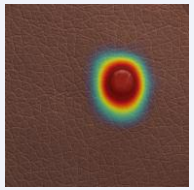
What this taught us

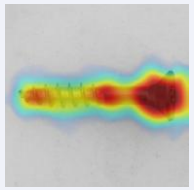
Weakest where defects are tiny or parts rotate: zipper 0.934, pill 0.881, capsule 0.841, grid 0.748, screw 0.565. Fine detail is lost at 224 px.

The threshold rule needs work: leather ranks perfectly (AUROC 1.000) yet 15 of 32 good samples were flagged. 20 images are too few to set a cut-off.

Heatmaps hold up: mean pixel AUROC 0.962, even where image scores are weak.


hazelnut


leather


screw (weak)

INT8 keeps accuracy, and directional light shows an early gain

Step 1: INT8 keeps accuracy (4 CPU threads)

PRODUCT	FP32 AUROC	INT8 AUROC	FP32 → INT8 MS
metal nut	0.985	**0.985**	21.4 → 14.3
screw	0.565	**0.568**	21.3 → 13.7
carpet	0.976	**0.972**	22.3 → 15.0
capsule	0.841	**0.838**	24.0 → 14.1

Model 11.1 MB → 2.9 MB. Calibrated only on defect-free metal-nut images, yet it holds on the other types.

Steps 2–4: lighting on Eyecandies

Light did not add up. Predicting the all-lights shot from single-light shots left 28–57% error: each render is tone-mapped. Our station must save linear RAW images to simulate light combinations.

Directional light helps (early signal): cropped to the candy at 320 px, every single directional light beat one fully lit shot, and combining them did best (chart below). At 224 px on the full frame the detector was near chance.

Image AUROC by lighting set-up (Eyecandies Lollipop, cropped, 320 px)



Early signal, not proof
50 test samples, 16 defective; only 3/16 caught at the threshold. The best pair was picked on the same samples.